

DIGITAL FORENSIC INVESTIGATION REPORT

Simplified Morris Worm – SEED/Docker Nano-Internet Laboratory

Case ID: CIP-B105-CS4_C11_26_DFIT_17305_NASIRU_LAWAL

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1. Executive Summary

The supplied evidence documents a controlled SEED/Docker nano-internet exercise involving a simplified Morris Worm training artifact. The lab contained 10.151.0.0/24, 10.152.0.0/24 and 10.153.0.0/24 Docker networks. The target host 10.151.0.71 was observed listening on TCP/9090. The worm source was configured to create a 500-byte badfile and transmit it to 10.151.0.71:9090. The generated badfile was recorded at 14:26:32.860796572 -0400 on 25 September 2026. Post-run evidence showed no worm.py process and no TCP/9090 listener on the VM. The laboratory containers were subsequently stopped and their final state preserved.

2. Investigation Objectives

- Document the lab topology and host baseline.
- Preserve and analyse worm.py and badfile.
- Compare pre- and post-run processes and network state.
- Construct a timeline from recorded timestamps.
- Document containment, cleanup and evidence integrity.
- Identify supported indicators of compromise.

3. Lab Environment

The analysis host was identified as VM running Linux 5.4.0-54-generic x86_64. Its primary address was 10.0.2.15/24. Docker bridge interfaces represented 10.100.0.0/24, 10.151.0.0/24, 10.152.0.0/24 and 10.153.0.0/24. The nano-internet included hosts, routers and an Internet-map container.

- Target: as151h-host_0-10.151.0.71; container ID 0930be2ff314
- Target service: ./server listening on 0.0.0.0:9090

- Observed lab networks: internet-nano_net_151_net0, _152_net0, _153_net0 and _ix_ix100

4. Evidence Inventory

The evidence directory contained baseline, post-run, timeline, Docker, artifact and integrity records, including worm_sha256.txt, worm_metadata.txt, badfile_metadata.txt, badfile_post_metadata.txt, process and network snapshots, Docker snapshots, evidence_manifest.sha256 and final_evidence_manifest.sha256.

5. Baseline Findings

The baseline date recorded in the transcript was 17:58:45 UTC on 25 September 2026. Baseline processes included normal Linux services, Docker/containerd activity and the nano-internet containers. Baseline listeners included TCP/22, 23, 21 and 8080 plus local DNS/printing-related listeners. Baseline connections also showed ordinary Firefox HTTPS traffic.

6. Worm Artifact Analysis

worm.py was 2,681 bytes, mode 0775, with SHA-256

e8a77eb10841d6fb60203819fd90f9fed3a0d3fca102e8c867ed18a79afdb4ab. Its recorded modification time was 2021-12-27 10:34:35 -0500. The source contains shellcode, creates a 500-byte badfile, targets 10.151.0.71, starts ping -q -i2 1.2.3.4, and transmits badfile using netcat to TCP/9090. The supplied sample also contains exit(0) after its attack cycle.

The source therefore supports an attempted payload-transmission capability. It does not, by itself, prove successful exploitation of the target.

7. Network and Target Evidence

The target container 10.151.0.71 was observed running ./server as PID 59 and listening on 0.0.0.0:9090. A TCP connectivity test from the VM reported success. At the inspection point, /worm was absent from the target container. The post-run VM snapshot contained no TCP/9090 listener and no active worm-related connection.

The available evidence supports interaction with the laboratory target service, but does not establish successful shellcode execution, persistence, or compromise of additional hosts.

8. Process Evidence

During timeline collection, PID 12305 ran `/bin/sh -c ping -q -i2 1.2.3.4` and PID 12307 ran `ping -q -i2 1.2.3.4`. Post-run searches for `worm.py` returned no `worm.py` process; `pgrep -af 'worm.py'` returned no result.

9. File and Timeline Analysis

The generated badfile was 500 bytes, owned by seed, mode 0664, and had modification/change time 2026-09-25 14:26:32.860796572 -0400. Its access time was 14:26:32.896795963 -0400. The directory listing showed `worm.py` and `badfile`, with `badfile` appearing after the exercise activity.

10. Timeline

Time	Event	Interpretation
17:58:45 UTC	Baseline captured	Initial host state documented.
16:49:16 UTC	10.151.0.71 inspected	Server PID 59 listening on TCP/9090; /worm absent.
14:26:32 -0400	badfile timestamp	500-byte payload artifact recorded.
14:26	ping process observed	Worm-associated visualization process recorded.
18:53:10 UTC	Cleanup timestamp	Containment/cleanup phase recorded.
18:53 UTC	Final Docker state	Lab containers recorded stopped, predominantly Exited (137).

11. Indicators of Compromise

- Labsetup/worm/worm.py SHA-256 e8a77eb10841d6fb60203819fd90f9fed3a0d3fca102e8c867ed18a79afdb4ab.
- Labsetup/worm/badfile 500 bytes; timestamp 2026-09-25 14:26:32.860796572 -0400.
- 10.151.0.71:9090/TCP configured worm target and observed laboratory service.
- Ping -q -i2 1.2.3.4 process observed during timeline collection.
- As151h-host_0-10.151.0.71 / container 0930be2ff314 target container.

12. Impact Assessment

Within the supplied evidence, demonstrated impact is confined to the controlled laboratory exercise. The evidence shows payload generation and an attempted transmission to the configured lab target. It does not establish successful remote code execution, persistence, or propagation to additional hosts.

13. Containment and Cleanup

Post-run evidence showed no worm.py process and no VM TCP/9090 listener. Cleanup was timestamped at 18:53:10 UTC. The final Docker snapshot recorded the main nano-internet containers and Internet-map container as exited, predominantly with status 137. The transcript also shows an attempted literal `kill <PID>` command that failed because no PID was substituted; the final Docker state nevertheless demonstrates that the lab containers were stopped.

14. Evidence Integrity

The transcript records generation of SHA-256 evidence manifests before and after cleanup. The worm source hash was explicitly recorded. The evidence set includes baseline, post-run and timeline artifacts suitable for before/after comparison.

15. Findings

- The lab topology was active and represented multiple Docker networks.
- 10.151.0.71 exposed TCP/9090 through ./server.
- worm.py was configured to create and transmit a 500-byte badfile to that service.
- badfile was created/modified during the recorded exercise activity.
- A worm-associated ping process was observed.
- No worm.py process remained in the post-run search.
- No TCP/9090 listener remained on the VM after the run.
- Successful exploitation or compromise of the target is not proven by the supplied evidence.
- The Docker lab was stopped and its final state preserved.

16. Recommendations

- Retain worm.py and badfile as evidence.
- Keep SHA-256 manifests with the submission.
- Use before/after snapshots as the primary forensic comparison.
- Describe the activity as attempted transmission unless exploitation is independently demonstrated.
- Avoid rerunning the worm unnecessarily; use static analysis and captured evidence for further validation.

17. Conclusion

The evidence documents a controlled Morris Worm training exercise in which worm.py generated its expected 500byte badfile and was configured to target 10.151.0.71:9090. The post-run evidence shows that the worm process was no longer active and that the Docker laboratory was subsequently stopped. The evidence supports an attempted payload transmission, but does not establish successful remote code execution or propagation to additional hosts.

Appendix Key Evidence Files

- evidence/baseline_date.txt
- evidence/baseline_hostname.txt
- evidence/baseline_ip.txt
- evidence/baseline_route.txt
- evidence/baseline_processes.txt
- evidence/baseline_process_tree.txt
- evidence/baseline_listeners.txt
- evidence/baseline_connections.txt
- evidence/worm_sha256.txt
- evidence/worm_metadata.txt
- evidence/badfile_metadata.txt
- evidence/badfile_post_metadata.txt
- evidence/post_process_tree.txt
- evidence/post_listeners.txt
- evidence/post_connections.txt
- evidence/post_docker_ps.txt
- evidence/post_docker_networks.txt
- evidence/timeline_file_times.txt
- evidence/timeline_processes.txt
- evidence/timeline_process_tree.txt
- evidence/evidence_manifest.sha256
- evidence/final_evidence_manifest.sha256
- evidence/cleanup_time.txt
- evidence/final_docker_state.txt